

ABDUL KALAM MUSTAQ

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PROFESSIONAL SUMMARY

GIS specialist with 5+ years of hands-on Esri ArcGIS experience across the full geospatial data stack, including data management, spatial and statistical analysis, and visualization. Currently working at the NJ Climate Change Resource Center, supporting build-out analyses for municipal CCRHVA Plans and the digitization of statewide hazard mitigation plans. Develops documented workflows, maintains QA/QC-governed spatial datasets, and automates analytical workflows using ArcGIS ModelBuilder and ArcPy. Skilled in ArcGIS Pro, ArcGIS Online, and StoryMaps, with experience producing metadata and methodology documentation that meets Center standards and communicating data-driven findings to research teams and external stakeholders.

TECHNICAL SKILLS

GIS Platforms (Esri): ArcGIS Pro, ArcGIS Online, ArcCatalog, ArcGIS Enterprise, Story Maps, Survey123, Hosted Feature Layers, Network Analyst, ModelBuilder
Analysis Toolkits: Spatial Analyst, Spatial Statistics, geostatistics, spatial autocorrelation, suitability & vulnerability analysis, raster / map algebra, build-out analysis, 3D analysis
Scripting & Automation: Python / ArcPy (automating GIS workflows), FME ETL, Census APIs, NJ Geocoder, SQL
Data Management & Governance: Spatial & tabular data management, metadata & methodology documentation, QA/QC protocols, geodatabase design, dataset publication, spatial databases
Statistics & Machine Learning: R, statistical analysis, machine learning (XGBoost, SMOTE, GridSearchCV) for spatial/predictive models, Power BI, Tableau, data visualization
Domain Data: Climate change, demographic (ACS/Census), environmental, planning, hazard mitigation, and critical-infrastructure datasets

PROFESSIONAL EXPERIENCE

- Graduate Research Assistant, New Jersey Climate Change Resource Center**
Data management, spatial analysis & visualization

New Brunswick, NJ
March 2026 to Present

 - Support data analysis and visualization for the NJ Climate Change Resource Center and its climate-impact programs, working with foundational spatial datasets that inform guidance and actionable science for stakeholders.
 - Lead dataset and workflow development for statewide **hazard mitigation planning**, digitizing Hazard Mitigation Plans across all 21 New Jersey counties among a team of full-time and graduate student staff.
 - Automated analytical workflows in **ModelBuilder** and **Python (ArcPy)**, including a build-out analysis pipeline that raised operational efficiency by 60% and accelerated delivery.
 - Developed and documented **methodologies, metadata, and QA/QC** procedures for county-wide datasets, ensuring data integrity and integration to Center and University publication standards.
 - Conducted Build-Out Analyses for 7 municipalities under multiple flood inundation scenarios (FEMA 1%, 1%+3 ft, 2 ft, 5 ft) to support Climate Change Hazard Vulnerability Assessments.
 - Built a **Survey123** data-collection solution integrated with Hosted Feature Layers to standardize a Hazard Mitigation Action Tracker, cutting manual data-entry errors by 95% and raising geocoding accuracy to 100%.
- Research & Development Analyst, Rutgers CLiME Lab**
Data manipulation, spatial analysis & statistics

Newark, NJ
January 2025 to September 2026

 - Engineered automated data pipelines (Python, Census APIs) that reduced processing time by 60%, and used **ArcGIS** to visualize housing and homelessness disparities across New Jersey.
 - Authored data-driven policy findings for a housing-affordability publication covering 12 New Jersey cities, presented to municipal planning officials.
- Urban Designer & Policy Advisor, Sponge Collaborative**
Design, Planning and Policy Implications, RFP's, Client and Project Management

Chennai, India
August 2021 to July 2024

 - Developed geospatial models in ArcGIS to identify 1,000 candidate sponge-park sites by ecological, social, and financial criteria; conducted cost-benefit analysis feeding \$12M in implementation funding.
 - Integrated BIM (Revit) and CAD-to-GIS for site selection across 122 candidate sites, streamlining the process by 45% and saving 70% of analysis labor.
 - Co-authored feasibility studies and institutional gap analyses supporting a \$9M National Disaster Management Authority grant; authored a "Sponge Parks" handbook adopted by Greater Chennai Corporation.

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SELECTED GIS PROJECTS

- **Urban Heat Vulnerability Analysis, City of Newark:** Multi-criteria raster (Spatial Analyst / map algebra) model combining land surface temperature and ACS demographics to identify populations most exposed to extreme heat.
- **Flood Vulnerability Mapping, Woodbridge Township:** ArcGIS sea-level-rise risk models identifying 163 at-risk residential parcels and 93.8% of critical wetlands under development pressure.
- **Crash Severity Prediction, NYC (Machine Learning):** XGBoost models on collision and weather data with SMOTE and GridSearchCV tuning, reaching a weighted F1 of 0.64.
- **Suitability & Network Analysis, 3D City Modeling:** ModelBuilder suitability modeling, ArcGIS Network Analyst datasets, and CityEngine procedural 3D visualization; geostatistics and spatial autocorrelation.

EDUCATION

Degree

Master of City and Regional Planning, Rutgers University, Bloustein School

Expected Dec 2026

Graduate Certificate in Geospatial Information Science and Technology

Bachelor of Architecture, First Class, Anna University, School of Architecture & Planning

May 2021

CERTIFICATIONS

Columbia University and Northeast Big Data Innovation Hub, 2025

National Student Data Corps (NSDC) - Transportation Data Science Program

IIT-Madras, UK-India-Brazil Winter School on Sustainable Urban Drainage Systems (SuDS), 2022

Professor Balaji Narasimhan, Department of Civil Engineering, IIT-Madras, 2022

Columbia GSAPP, 2024

Affirmations Fall and Spring 2023-24 cohort, curated by Andrés Jaque and Bart-Jan Polman.

COMPETITION HONORS

- Winner, Chennai Lakefront Rejuvenation Competition, Chennai Metropolitan Development Authority (2023).
- Nominated, National Awards for Excellence in Architectural Thesis, Council of Architecture, India (2022).